Inside a remoll ROOT file

Particles, tracks and detector records.

T contains · rate · ev · bm · part[] · hit[] · sum[] · units · seed · dets

run_data
separate object · one run

One T entry describes an event. Tracks belong to that event; multiple hit records can belong to one track. **Track IDs restart for each event and may be reused in other events.**

Event & beam

rate · ev · bm

Event kinematics and target-sampler state, stored once per event.

rate — Hz
This history's rate weight, normally cross section × effective luminosity / Nthrown. Every hit in the event shares this event weight.
src/remollIO.cc:171
src/remollPrimaryGeneratorAction.cc:212

ev.A · ev.Am — ppb
A is the physics asymmetry; Am includes beam polarization. Both are already numeric ppb.
src/remollEvent.cc:103–104

ev.xs — μb
Effective cross section in microbarns. moller already includes phase space and target atomic number Z, with 1/2 for identical-electron double coverage.
src/remollEvent.cc:102
src/remollGenMoller.cc:44

ev.Q2 · ev.W2 · ev.thcom
Q2 is momentum transfer squared; W2 is hadronic invariant mass squared (MeV²). moller writes Q2 = 0 as a placeholder and W2 stays at its unset sentinel. thcom (rad) depends on the generator: Møller's first-electron COM angle (partner π − thcom); Al/C12's lab angle.
src/remollEvent.cc:105–107,157 · src/remollGenMoller.cc:55,59 · src/remollGenAl.cc:101 · src/remollGenC12.cc:103

ev.beamp — MeV
Generator beam-momentum magnitude. moller approximates |p| with sampled total beam energy; beam and pElastic store exact momentum.
src/remollEvent.cc:108 · src/remollVEventGen.cc:91 · src/remollGenpElastic.cc:180 · src/remollGenBeam.cc:198

bm.x · bm.y · bm.z
bm.dx · bm.dy · bm.dz
bm.th · bm.ph
Target-sampler interaction position (x/y/z, mm), beam direction (dx/dy/dz, normally unit length) and angles (th/ph, rad). beam/external bypass this sampling: bm may be default or stale. part[] records the actual primary launch state.
include/remollBeamTarget.hh:134–141 · src/remollBeamTarget.cc:219,225 · src/remollGenBeam.cc:39,200 · src/remollGenExternal.cc:33

Generated primaries

part[]

Initial particle records; secondaries do not get entries in part[].

part.pid
PDG species code of a generated primary: 11 electron, −11 positron, 22 photon.
src/remollEvent.cc:51

part.trid
remoll writes the primary's array index + 1. The standard moller pair has part.trid 1 and 2, both Geant4 primaries. part[] has no transport secondaries; hit.trid labels whichever transported track made a hit.
src/remollEvent.cc:67,80

part.vx · part.vy · part.vz — mm
Initial global position supplied to transport. Both generated Møller electrons start at the same sampled interaction point.
src/remollEvent.cc:55–57
src/remollVEventGen.cc:143

part.px · part.py · part.pz · part.p — MeV
Initial lab momentum components and magnitude, after any beam/scattering rotations. The hit fields describe the later transported state.
src/remollEvent.cc:58–60,63

part.th · part.ph — rad
Polar angle and azimuth of the primary's initial lab momentum.
src/remollEvent.cc:61–62

part.tpx · part.tpy · part.tpz — MeV
The generator's original momentum components, before beam-direction and multiple-scattering rotations.
src/remollEvent.cc:64–66

part.sx · part.sy · part.sz — dimensionless
For standard moller generation this stored polarization is zero. The gun separately polarizes only primary 1 when configured; hit.sx/sy/sz record transported polarization.
src/remollEvent.cc:52–54
include/remollEvent.hh:29
src/remollPrimaryGeneratorAction.cc:187

part.tjx · part.tjy · part.tjz — mm arrays
Aligned arrays of global positions along a saved primary trajectory. They are empty when no trajectory was stored.
src/remollEvent.cc:84–86

Deposited-energy sums

sum[]

One subtotal per detector and copy, before individual-hit filtering.

sum.det · sum.vid
Sensitive-detector number and physical-volume copy number for this subtotal.
include/remollGenericDetectorSum.hh:41–42

sum.edep · sum.n
Deposited energy (edep, MeV) and step count (n), including zero-deposition steps. Individual-hit filters can omit hit[] records while those steps still contribute to the sum.
include/remollGenericDetectorSum.hh:43–44

sum.by_pid · sum.by_pid.pid · sum.by_pid.n
sum.by_pid.edep
One subtotal per PDG species, carrying its code (pid), step count (n) and deposited energy (edep).
include/remollGenericDetectorSum.hh:48–49
src/remollGenericDetectorSum.cc:33–34

sum.by_pid.x · sum.by_pid.y · sum.by_pid.z — mm
Energy-weighted global position of that species' deposit; meaningful only when its edep > 0.
src/remollGenericDetectorSum.cc:38–40

Detector hit records

hit[]

Each entry describes a recorded sensitive-detector step. One track can leave many entries.

IDENTITY

hit.det
The sensitive-detector number assigned by remoll's geometry setup.
include/remollGenericDetectorHit.hh:64

hit.id
Normally the pre-step physical volume's copy number from Geant4's touchable history. Legacy GEM reconstruction (detectors 501–504) can change it when different tracks hit one detector.
include/remollGenericDetectorHit.hh:65 · src/remollGenericDetector.cc:154 · src/remollEventAction.cc:70 · src/remollTrackReconstruct.cc:35,42,45,314

hit.pid
The PDG species code: electron 11, positron −11, photon 22; μ^- 13, μ^+ −13.
Pions: π^0 111, π^+ 211, π^- −211; neutron 2112, proton 2212.
Kaons: K^+ 321, K^- −321, K^0_L 130, K^0_S 310. Geant4 optical photons have code −22.
include/remollGenericDetectorHit.hh:68 · PDG numbering (2024), pp.1,4–7 · Geant4 G4OpticalPhoton.cc:67

hit.trid
Track ID follows a particle's evolving Geant4 position, momentum, energy and time across its hits. IDs begin at 1 in each event. For standard moller generation, 1 and 2 are the outgoing electrons; either, both or neither may reach a detector.
include/remollGenericDetectorHit.hh:66
src/remollGenericDetector.cc:308

hit.mtrid
Mother track ID is remoll's name for Geant4's ParentID; 0 denotes a primary. Both electrons from the built-in moller generator have 0. Their direct children have mtrid 1 or 2; later descendants name their immediate parent.
include/remollGenericDetectorHit.hh:67
src/remollGenericDetector.cc:309

hit.gen
Legacy integer cast of the creator-process pointer. Both generated Møller primaries have gen = 0; this is not a stable process code. Parentage is carried by mtrid.
include/remollGenericDetectorHit.hh:69
src/remollGenericDetector.cc:312

POSITION & TIME

hit.x · hit.y · hit.z — mm
Hit position in the global/world frame. Normally this is the start of the recorded step (pre-step point). An optical photon ending on a boundary can instead record the post-step point. This locates the encounter, not the particle's birth.
include/remollGenericDetectorHit.hh:71–73
src/remollGenericDetector.cc:273,287

hit.xl · hit.yl · hit.zl — mm
The same selected point translated and rotated into that point's touched-volume frame. The origin and axes belong to that volume, so different tiles need not share one frame.
include/remollGenericDetectorHit.hh:74–76

hit.r — mm
Global cylindrical radius, $r = \sqrt{x^2 + y^2}$: distance from the world z-axis. It is not the distance from a tile's center or from an offset beam.
include/remollGenericDetectorHit.hh:77

hit.ph — rad
Azimuth of the global hit position, atan2(y, x). It describes where the hit lies around the z-axis. A momentum azimuth would instead come from atan2(py, px).
include/remollGenericDetectorHit.hh:78

hit.vx · hit.vy · hit.vz — mm
The global position where this track began, copied from Geant4's track vertex. It remains attached to the track as it travels. A secondary's vertex can lie in material far from the beam–target interaction.
include/remollGenericDetectorHit.hh:88–90

hit.t
Geant4 global time at the recorded point, measured from the start of the event. The same pre/post-step choice determines position and time; this is not a wall-clock timestamp.
include/remollGenericDetectorHit.hh:70

TRACK STATE

hit.px · hit.py · hit.pz — MeV
Current track momentum in the world axes when the hit is recorded. Its direction can change between records. Positive pz means motion toward +z and negative pz toward −z, regardless of ancestry.
include/remollGenericDetectorHit.hh:79–81

hit.pxl · hit.pyl · hit.pzl — MeV
That same momentum rotated into the touched volume's local axes. Only the rotation applies: moving the coordinate origin does not change a momentum vector.
include/remollGenericDetectorHit.hh:82–84

hit.p — MeV, c = 1
Magnitude of the current track momentum, |p|. part.p describes a generated primary's initial lab momentum; hit.p describes the transported track at this recorded encounter.
include/remollGenericDetectorHit.hh:91

hit.e · hit.k — MeV
e is total energy and k is kinetic energy. For a massive particle, e includes rest energy and k does not; for a photon, e = k. Neither is the energy deposited in the detector, which is hit.edep.
include/remollGenericDetectorHit.hh:92,94

hit.m — MeV, c = 1
The rest mass of the track's particle species, copied from its Geant4 particle definition. It is distinct from both the current total and kinetic energies.
include/remollGenericDetectorHit.hh:93

hit.beta — v / c
Current track speed divided by the speed of light. It is dimensionless; values near 1 describe relativistic motion.
include/remollGenericDetectorHit.hh:95

hit.sx · hit.sy · hit.sz — dimensionless
The track's current polarization vector from Geant4. These dimensionless components are not spin angular momentum in units of ħ; their interpretation depends on the particle and physics model.
include/remollGenericDetectorHit.hh:85–87

hit.edep — MeV
Energy deposited during this recorded step, rather than energy still carried by the particle. Stored hits may omit filtered steps, while sum.edep accumulates the detector copy's contributing steps.
include/remollGenericDetectorHit.hh:96

Built-in Møller event

Two outgoing e− primaries; example hits and descendants.

OUTGOING e− 1
part.trid = 1
hit A: trid 1, mtrid 0
hit B: trid 1, mtrid 0
two recorded hits

OUTGOING e− 2
part.trid = 2
hit C: trid 2, mtrid 0
also has mtrid = 0

during transport

TRANSPORT SECONDARY
hit.trid = 3
hit.mtrid = 1
no part[] entry

creates

NEXT GENERATION
hit.trid = 4
hit.mtrid = 3
no part[] entry

The hard scatter is generated before transport. Both outgoing e− enter Geant4 as primaries: pid = 11, mtrid = 0. IDs 1 and 2 follow generator order, not a fixed energy ranking.
src/remollGenMoller.cc:64–70
src/remollPrimaryGeneratorAction.cc:175–199
src/remollEvent.cc:67 · src/remollGenericDetector.cc:308–309
Geant4 G4ParticleGun.cc:59,224–238
Geant4 G4EventManager.cc:412–415,468
Geant4 G4PrimaryTransformer.cc:256–262 (11.4.2)

Run information

seed · dets · run_data

seed and dets are in T; run_data is a separate object for the run.

seed.fRunNo · seed.fEvtNo · seed.fSeed
Run number, event number and serialized random-engine state. fSeed is state text, distinct from the numeric run seed.
include/remollIO.hh:46–48

dets.sd · dets.lv
Intended name-to-number leaves for sensitive detectors (sd) and logical volumes (lv). This revision passes vector-object addresses to ROOT's leaf-list writer, so the saved numbers are unreliable.
src/remollIO.cc:88–89
include/remollIO.hh:144,146

run_data.fNthrown · run_data.fSeed
fNthrown is the requested event count. fSeed records the numeric /remoll/seed value; it stays 0 when only startup or -r seeds Geant4.
include/remollRunData.hh:47,53
src/remollRunData.cc:18
src/remollRunAction.cc:37
remoll.cc:149

run_data.fGitInfo · run_data.fRunTime
run_data.fRunPath · run_data.fHostName
Build revision, run timestamp, working directory and host machine name.
include/remollRunData.hh:59,64,69,74

run_data.fMacro · run_data.fGDMLFiles
run_data.fMagData
Saved run macro, GDML texts and field-map records: the inputs recorded with the run.
include/remollRunData.hh:29,34,39

filedata.filename · filedata.hashsum
filedata.timestamp
Field-map filename and modification timestamp. hashsum is the literal "no hash" in this revision; no checksum is computed.
src/remollGlobalField.cc:210–211,219

remollTextFile.fFilename
remollTextFile.fFilenameSize
remollTextFile.fBuffer
remollTextFile.fBufferSize
Each saved text has a filename and content buffer with their sizes. Both sizes include the terminating NUL.
src/remollTextFile.cc:75–76,79–80

Unit factors

units

Internal units: mm, MeV, ns and rad. Helpers are numerical scale factors.

units.nm · units.um · units.mm · units.cm
units.m
In mm: nm = 10^{-6}, um = 10^{-3}, mm = 1, cm = 10, m = 1000.
src/remollSystemOfUnits.cc:9–13

units.mm2 · units.cm2 · units.m2 · units.mm3
units.cm3 · units.m3
Area: 1, 10^2, 10^6 mm². Volume: 1, 10^3, 10^9 mm³.
src/remollSystemOfUnits.cc:15–17,19–21

units.eV · units.keV · units.MeV · units.GeV
In MeV: eV = 10^{-6}, keV = 10^{-3}, MeV = 1, GeV = 1000.
src/remollSystemOfUnits.cc:23–26

units.rad · units.deg · units.sr
rad = 1; deg = π/180 rad; sr = 1 steradian.
src/remollSystemOfUnits.cc:28–30

units.barn · units.mbarn · units.ubarn
1 barn = 10^{-22} mm²; mbarn = 10^{-25}; ubarn = 10^{-28}. ev.xs is already numeric microbarns.
src/remollSystemOfUnits.cc:32–34

units.nsec · units.msec · units.sec
In ns: nsec = 1, msec = 10^6, sec = 10^9.
src/remollSystemOfUnits.cc:36–38

units.ppm · units.ppb
ppm = 10^{-6}; ppb = 10^{-9}. ev.A and ev.Am are already numeric ppb.
src/remollSystemOfUnits.cc:6–7

units.Hz · units.kHz · units.MHz · units.GHz
As implemented: Hz = 1/s; kHz = Hz/10^3; MHz = Hz/10^6; GHz = Hz/10^9. rate is already numeric Hz.
src/remollSystemOfUnits.cc:40–43